RIPHAH
INTERNATIONAL UNIVERSITY

FACULTY OF COMPUTING

SE – 3

ASSIGNMENT - 01

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SUBJECT: Coal Lab

Assembler: An assembler is a system software that converts assembly language into machine code.
Compiler: A compiler is a system software that translates the entire high-level language program into machine code at once.
Interpreter: An interpreter is a system software that translates and executes a high-level language program line by line.

Basis of Comparison	Assembler	Compiler	Interpreter
Definition	Translates assembly language into machine code	Translates high-level language into machine code	Translates and executes high-level language line by line
Input Language	Assembly language	High-level language	High-level language
Output	Machine code	Object code / executable file	No separate object code
Translation Method	One-to-one instruction translation	Entire program translated at once	Statement by statement
Error Detection	Errors shown after assembling	Errors shown after compilation	Errors shown during execution
Execution Speed	Very fast	Fast	Slow
Memory Requirement	Low	More than assembler	Less than compiler
Portability	Machine dependent	Machine dependent	Machine independent
Examples	MASM, NASM	C, C++, Go	Python, JavaScript